

006051

REPORT OF EXAMINATION

of the

GABERO MINE,

JORDAN RIVER, B.C.

Property File

092C 073

Sunro

To:

F. R. Weekes, Manager,
Persupine Goldfields D & F. Co. Ltd.,
Montreal, Que.

By:

Chas. C. Starr
O. D. Frith.

Nelson, B. C.
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INTRODUCTION:

In all, five days were spent in examining this property. Camp was made with the Vancouver Island Power Co., at their No. 2 camp, and due to the exceptionally rough topography of the country, considerable time was taken up, in reaching and finding the various workings.

DISTRICT & LOCATION:

The Gabbro group is located in the Jordan River Section of the Victoria Mining Division. It adjoins the Sunloch group on the Jordan River, about 2 miles from the mouth, where the river empties into the Straits of Juan de Fuca.

PROPERTY:

The group consists of twenty-three claims or fractions. These claims are staked in an irregular block, crossing over the Jordan River, and tying to the Sunloch group on the west side, (see map).

The property is owned by the Gabbro Copper Mines Ltd., Sayward Building, Vancouver, B. C. Mr. Geo. E. Winkler is Managing Director.

PRICE & TERMS:

No definite terms have been arrived at.

TRANSPORTATION:

A good motor road runs from Victoria, a distance of 42 miles, to the Vancouver Island Power Co's plant which is about 3 miles from the property. From here the Sunloch Co. have a narrow gauge railway to their tunnels. This follows the Jordan River on the south side, cutting

through the Gabbro property, and is largely, cut out of the solid rock. From the Sunloch tunnels, the workings on the Gabbro claims are reached, by very steep and treacherous switchback trails.

It is unlikely that, the Sunloch railway could be used, if the two properties were not worked under one head, but a new railway could be built to the sea level. Due to the extremely heavy rock work, this would be expensive, but along the steep banks of the Jordan River is the only place that a road or railroad could be constructed to advantage.

An aerial tram way could be constructed but it is possible that power may be required to drive it.

Ore would be shipped by boat, from or near the mouth of the Jordan River, to any sea point.

POWER, WATER, WOOD & CLIMATE:

The property is only three miles from the Vancouver Island Power Co. plant, and abundant electric power could be secured, at a reasonable rate.

Several small creeks cross the property flowing into Jordan River, where a good supply of water could be developed.

There is a good growth of timber on the property, suitable for both building and mining purposes. It is likely that the timber is owned by some timber company, as logging operations were noticed in the vicinity.

There is very little snow in the winter, but the rain fall is exceptionally heavy.

TOPOGRAPHY:

The Jordan River flows through a very steep canyon, whose walls average about 60 degrees for a vertical distance of 600 feet, with cliffs in many places. Above this on each side of the river the ground has a gradual slope. Towards the west the property slopes off gradually to the sea except for several small creek basins.

HISTORY:

The Gabbro and Sunloch groups were originally staked by Geo. E. Winkler, of Victoria, B. C., as one group, who did assessment work on them.

In the spring of 1917, the whole group was bonded to R. H. Stewart and associates, of Vancouver, who organized the Sunloch Mining Co. Ltd.

At some period between 1917 and 1919, the two groups were bonded to the Consolidated Mining and Smelting Co. The Gabbro group was dropped by this Company, and a majority of stock in the Sunloch Co. purchased.

The Gabbro Copper Mines Co. Ltd. were then organized, with Geo. E. Winkler as Managing Director.

The property was later bonded by Augustus Locke and associates, of San Francisco, Cal. After some surface work the bond was relinquished.

BIBLIOGRAPHY:

Very complete reports are to be found in the Minister of Mines Reports on the District, from 1917 to 1925, as follows:

1917: Report on work and geology of Sunloch, with sketch map.

- 1918: Report on work and geology by R. H. Stewart, with map.
- 1919: Report of Sunloch with map. Short note re. Gabbro.
- 1920: Short Description of Geology and work on the Gabbro.
- 1921: Report of development on the Gabbro, with good map of surface Development and geological and topographical map.
- 1922: Extensive report on the Gabbro, with map showing claims on Jordan River.
- 1923: Report of Gabbro, with assays.
- 1924: Report, ineffectivity.
- 1925: Report, Crown Grants obtained on most promising claims.

Memoir 13 Southern Vancouver Island, Geological survey, covers this district, but is not accurate.

No. 80 Geological Series, Sooke and Duncan Map Area, covers adjacent ground, but does not cover the area in which the Gabbro is located.

DEVELOPMENT:

Development has been done by stripping and trenching principally. A small tunnel was driven on what is known as the Winkler zone for about 30 feet, and a short tunnel on the Hornet zone.

Stripping which has been done, is confined principally to the steep side hill and cliffs, where there is little overburden. In many places, it has not broken through the oxides.

GEOLOGY:

The rocks underlying the property are of Tertiary age and consist of the Hatcherian volcanics, largely fine grained basalts, which are intruded by stocks of Sooke Gabbro.

The approximate contacts of the gabbro are shown on the claim map as worked out by Mr. Winkler. These contacts were not checked, except at one or two points, but they are thought to be approximately correct; since the actual contact is nearly always covered with soil no great detail can be obtained.

The basalts are mostly hard compact flows, but there are occasional beds of tuffaceous material, and the whole has been strongly metamorphosed with the development of hornblende.

The gabbro is a massive dark gray to green rock and shows considerable magmatic differentiation. Both the gabbro and the basalts, especially near the contacts and in the shear zones, are intruded by tiny aplitic dikes which are often very numerous and not more than a fraction of an inch across. The gabbro also contains many inclusions of basalt, some of them being of considerable size.

Numerous, but not very intense, zones of shearing cut both the basalts and the gabbro, most of them running approximately parallel to the contacts. So far as known they do not cut the contacts. These shear zones are more highly metamorphosed than the remainder of the rock, probably on account of the aplitic seams which are most highly developed in them, and are mineralized with pyrite, chalcopyrite and pyrrhotite, and have been somewhat silicified although little quartz is to be seen.

The metallic sulphides occur both as disseminations through the rock, and in seams and veinlets roughly parallel to the shearing. Generally the walls of the shear zones are indefinite and not very continuous.

SAMPLING:

(See sample maps)

Channel samples were taken with moils.

A total of 22 samples were secured, as follows:

Caulfield zone 13, Cave zone 6, and the Turnbull zone 3.

The best assay obtained (Sample 1074) shows 4.7% copper. This was massive chalcopyrite but apparently does not continue the same grade in either direction.

Samples 1085 and 1087 contained considerable oxides, while the balance were fairly clean.

Between samples 1090 and 1091 there was a strip of waste 18" wide, which was not sampled.

Sample 1077 was taken on a narrow cross stringer of ore.

Omitting the high grade samples (1074 and 1087) the numerical average of the balance, is 0.78% copper.

DESCRIPTION OF ZONES:

The Cave zone has been rather extensively developed on the Sunloch property east of the river, and what appears to be the same zone extends across the northeast corner of the Vulcan No. 5 claim, striking about N 60 W and dipping about vertical.

At the western end of the exposures there is a streak of fairly good ore, but most of the cuts show very low grade material. Walls are poor or indistinguishable, and the ore of a workable grade is too narrow to be commercial. According to the Minister of Mines Report, 1917, the ore in the Cave tunnel of the Sunloch averages about 2.4% copper.

The Turnbull zone is about 150 feet to the southwest of the Cave zone and runs about parallel to it. Its

copper content is consistently too low to be classed as ore at the seven points where it has been prospected. It is about four feet wide.

The Caulfield zone is on the Black Hornet claim, in the basalt on the south side of the gabbro, a couple of hundred feet from the contact. Its strike is approximately S 60° W and the dip 45° north; the hanging wall is generally well defined and the mineralized zone is fifteen to twenty feet wide, but the better part of the mineralization is about half that. Near the east line of the Black Hornet claim a northeast striking shear enters it from the south.

The Winkler zone is on the Vulcan No. 2 claim and strikes in a northeasterly direction. A short tunnel has been driven on it near its intersection with a northwesterly shear, and is said to show several feet of low grade ore.

It lies in the Gabbro, not far from the south contact. The Stewart and Pat zones also lie in the Gabbro several hundred feet from, and more or less parallel, to the contact. They are opened by several shallow cuts and appear fairly well defined and of fair size, but do not show any commercial ore.

EFFECT OF SUNLOCH DEVELOPMENT ON GABBRO PROPERTY:

The portal of the Cave tunnel is about 600 feet from the Gabbro boundary to the northwest along the strike of the zone, and the breast of the tunnel is 400 feet from the Gabbro ground (Gabbro claim) to the south east at a depth of about 300 feet. No definite average value is given for the ore in this tunnel, but it is to be inferred from the Minister of Mines reports and other sources that it is about

21% copper. To the southeast, the zone is covered on the Gabbro claim but there is no reason to doubt the existence of similar ore there. To the northwest the zone is strong on the Vulcan No. 5 and 6 claims, but the surface samples indicate lower grade ore, or no ore.

To the northward the River zone will not enter the Gabbro property, and the Archibald zone has not been identified there. To the southward the River zone has been developed to within 300 feet of the Gabbro line, at a depth of some 600 feet and shows 21% copper at the face. This ore, if it continues on the same strike, should enter the Gabbro ground near its sideline, but the zone has not been found there, probably on account of being covered.

CONCLUSION:

The various ore zones have merely been scratched on the surface on the Gabbro group and there is ample space remaining for good shoots of ore to be found. The samples, however, which were taken on the Cave, Turnbull, and Caulfield zones, show material of generally too low grade and too narrow widths to form commercial ore in the present exposures. Admitting that the surface samples are somewhat low on account of oxidation and leaching, it is still improbable that the unaltered material will be sufficiently higher to be interesting.

The Winkler, Stewart and Pat zones were not sampled but they are not cloized, nor do they look, to have as good ore as the ones that were sampled.

Such as they are, the ore zones may reasonably be expected to hold their size and values to considerable

depths, and they are in several instances proved to have a considerable length.

It is entirely probable that exploration on the Gabbro claim, on the strike of the River and Cave zones, will show ore of the approximate size and value as that opened in the Sunloch, which according to the report of the Resident Engineer is sufficient to make pay ore. However, to make the proposition attractive there must be a very large tonnage of ore and cheap working costs.

The Sunloch and Gabbro groups should be worked as one property, both for topographical reasons, and because otherwise the Sunloch No. 6 claim cuts the Gabbro property into two parts, in so far as the River, Archibald, Cave, and Turnbull zones are concerned and conversely, the Gabbro contains the extensions of these zones, leaving a limited area of promising ground for the Sunloch. Under such conditions the total cost of equipping and operating the properties as two units will be very considerably higher per ton than if they were one unit. If both groups were offered for sale they would make a property that would be only moderately attractive, assuming that the workings on the Sunloch have ore of the size and value reported.

Since the Gabbro Group must be considered by itself, it is distinctly unattractive on account of the low grade of the known ore and its position relative to the Sunloch.

Respectfully submitted,

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